

Comptia A+ Assessment

[Section A — Concept Application]

- ❖ **SCRENARIO:** - You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.
- ❖ **Question:** - Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.
 - To select compatible components for an office workstation, I would follow these steps:
 - 1. Select the Motherboard:
 - First, choose a motherboard with the required CPU socket, RAM type, storage interfaces, and expansion slots.
 - 2. Select the CPU:
 - Choose a CPU that is specifically compatible with the motherboard's CPU socket and supported CPU family.
 - 3. Select the RAM:
 - Choose RAM that matches the motherboard's supported DDR generation, capacity, speed, and number of slots.
 - 4. Select the Storage Device:

- Choose an SSD or HDD that matches the motherboard's available SATA or M.2/NVMe interface and provides sufficient capacity for office work.

➤ 5. Confirm Compatibility:

- Before purchasing, I would compare the motherboard manufacturer's specifications with the CPU, RAM, and storage specifications. I would confirm that the CPU socket, RAM type, and storage interface are supported by the motherboard.

❖ **SCRENARIO: - You are advising a client whose employees mainly run office and productivity applications on their workstations.**

❖ **Question: - Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.**

➤ HDD (Hard Disk Drive):

- HDD stores data on spinning magnetic disks. It provides large storage capacity at a lower cost but has slower read/write speeds because it contains moving mechanical parts.

➤ SSD (Solid-State Drive):

- SSD uses flash memory and has no moving parts. It is much faster, quieter, and more reliable than an HDD, making applications and system startup faster.

➤ NVMe:

- NVMe is a high-speed type of SSD that uses the PCIe interface instead of the traditional SATA interface. It provides very fast data transfer and low latency.

➤ Recommendation

- For employees mainly using office and productivity applications, I would recommend an SSD, preferably an NVMe SSD if the budget and motherboard support it. An SSD provides fast boot times, quick application loading, and better overall responsiveness than an HDD.

➤ Technical Comparison:

Storage	Technology	Speed	Best Feature
HDD	Magnetic disk	Slow	Low cost, high capacity
SSD	Flash memory	Fast	Good overall performance
NVMe	Flash + PCIe	Very fast	High performance and low latency

- ❖ **SCRENARIO: - You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.**
- ❖ **Question: - Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.**

➤ GPU (Graphics Processing Unit):

- The GPU processes graphics and visual workloads generated by design software. A suitable GPU provides smooth rendering and helps maintain good performance when graphics-intensive applications are used for long periods.

➤ Cooling System:

- The cooling system removes heat produced by the CPU and GPU during continuous operation. It uses components such as fans and heatsinks to maintain safe operating temperatures.

➤ Why Both Are Important:

- A capable GPU provides consistent graphics and rendering performance.
- Effective cooling prevents excessive heat buildup.
- Proper cooling helps avoid thermal throttling, performance loss, and unexpected shutdowns.
- Together, they help maintain system performance and stability during sustained workloads.

❖ **SCRENARIO: - You are assembling a PC from components sourced from three different vendors.**

❖ **Question: - Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.**

➤ Hardware compatibility is important because all computer components must work together according to their supported specifications and interfaces. Before assembly, I would check the motherboard's CPU socket, RAM type, storage interfaces, expansion slots, and PSU requirements.

➤ Problems Caused by Incompatible Components:

1. **CPU and Motherboard:** An incompatible CPU socket may prevent the processor from fitting or working with the motherboard.
2. **RAM:** Using an unsupported RAM type or generation may cause the system not to boot or detect the memory correctly.
3. **Storage:** An incompatible SATA/M.2 interface may prevent the storage device from being detected.
4. **GPU:** An unsuitable slot, power connection, or insufficient PSU capacity can cause display or stability problems.
5. **PSU:** An incorrectly rated PSU may not provide enough power, causing crashes, shutdowns, or failure to start.

❖ **SCRENARIO: - You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.**

❖ **Question: - Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.**

- I would choose a PSU based on the total power requirement of the CPU, GPU, storage devices, fans, and other components, with some additional capacity for safe operation and future upgrades. I would also select a reliable PSU with the required power connectors and appropriate efficiency rating.
- Safety and Connection Checks:
- 1. Check PSU Rating:
 - Confirm that the PSU provides enough wattage for the complete system and has the correct 24-pin motherboard, CPU power, GPU PCIe, and SATA power connectors.
- 2. Switch Off and Unplug Power:
 - Before making connections, ensure the PSU is switched off and the AC power cable is disconnected.
- 3. Connect the Motherboard:
 - Firmly connect the 24-pin ATX motherboard power connector and the required CPU power connector.
- 4. Connect the GPU:

- If the graphics card requires additional power, connect the appropriate PCIe power connector securely.

➤ 5. Connect Storage Devices:

- For SATA drives, connect both the SATA data cable to the motherboard and SATA power cable from the PSU.

➤ 6. Final Safety Check:

- Check that all connectors are fully seated, cables are clear of fans, and there are no loose screws or exposed connections inside the case.

❖ **SCENARIO: - You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.**

❖ **Question: - Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.**

- After assembling the workstation, I would perform the following checks to confirm that the system passes POST (Power-On Self-Test) successfully.

➤ 1. Check Power Connections:

- Turn on the system and confirm that the PSU, 24-pin motherboard connector, CPU power connector, and GPU power connector are properly connected.

➤ I would check for:

- Fans spinning, power LEDs, and other normal signs that the system is receiving power.

➤ 2. Observe the POST Process:

- Power on the PC and observe the startup screen.

➤ I would check for:

- Manufacturer logo or POST screen.
- No POST error message.
- No abnormal beep/error code.
- Keyboard responding to BIOS/UEFI input.

➤ 3. Check Hardware Detection in BIOS/UEFI:

- Enter the BIOS/UEFI setup and verify that the main hardware is detected.

➤ I would confirm:

- CPU is detected.
- Correct amount of RAM is detected.
- SSD/HDD is detected.
- GPU/display output is working where applicable.

➤ 4. Confirm Successful Boot:

- Allow the system to continue from POST into the operating system.

➤ Expected result:

- The computer should boot normally without POST errors, hardware detection warnings, or unexpected shutdowns.

➤ If POST Fails:

- If the system does not pass POST, I would troubleshoot in a systematic order:

- **Power:** Recheck PSU and motherboard/CPU/GPU power connections.
- **RAM:** Reseat the RAM and test one module at a time if multiple modules are installed.
- **GPU/Display:** Check GPU seating, GPU power and display cable connections.
- **Storage:** Check SATA/M.2 connections and confirm the drive is detected in BIOS.
- **POST Codes/Beep Codes:** Note any beep or diagnostic code and use it to identify the suspected hardware problem.
- **Clear CMOS:** If appropriate, reset BIOS settings and then recheck the hardware configuration.

[Section B — Practical Coding Tasks]

❖ Task 1: Component Identification & Functions

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table.

➤ **Ans:** - Identify the main hardware components of a desktop PC and document their functions and visible specifications.



➤ Components Identified:

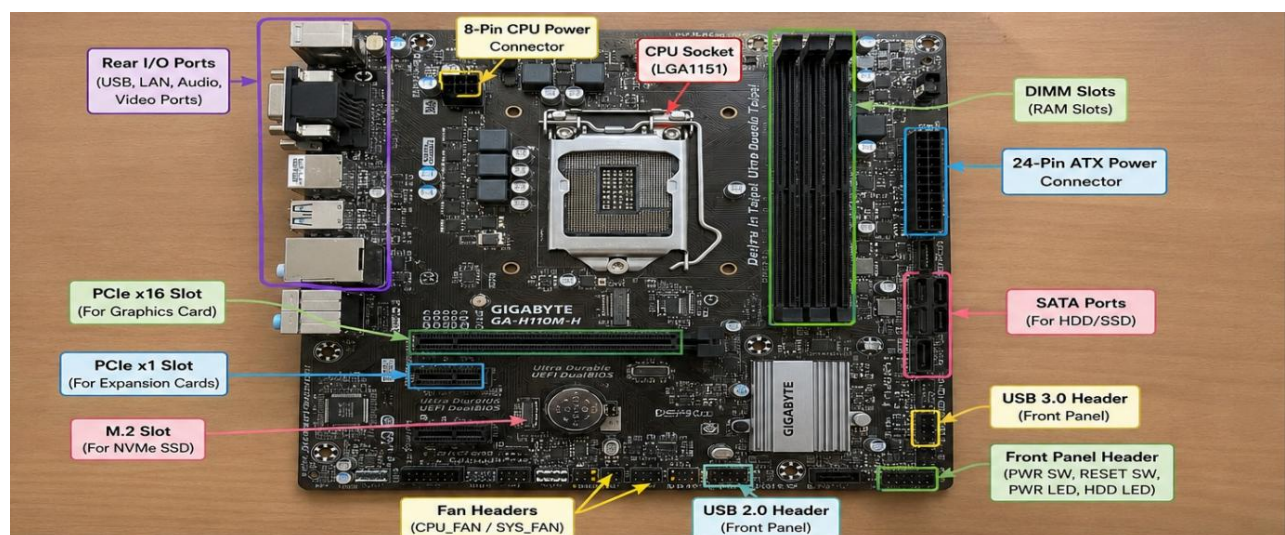
Component	Main Function	Make/Model or Key Specification
Motherboard	Connects and allows communication between the CPU, RAM, storage, GPU and other hardware components.	Record the motherboard make/model visible on the board.
CPU (Processor)	Performs calculations and executes instructions required to run the operating system and applications.	Record CPU model/name if visible.
RAM (Memory)	Temporarily stores data and instructions that are actively being used by the CPU, allowing quick access.	Record DDR type and capacity, e.g. DDR4/DDR5 and 8 GB/16 GB.
GPU (Graphics Card)	Processes graphics and sends visual output to the monitor. It is especially useful for gaming and graphics-intensive applications.	Record GPU make/model and VRAM if visible.
Storage Device	Permanently stores the operating system, applications, documents and other user data.	Record type (SATA SSD/NVMe SSD/HDD) and capacity if visible.

Cooling System	Removes heat from components such as the CPU and GPU and helps maintain safe operating temperatures.	Record cooler/fan type or model if visible.
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❖ Task 2: Motherboard Ports & Socket Identification

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).
- Identify all expansion slots (PCIe, M.2, etc.) and their purpose.
- Identify the CPU socket type and note which CPU families are compatible with it.
- Photograph or diagram the motherboard with labels.

➤ **Ans:** - To identify the major ports, expansion slots, and CPU socket on a motherboard and understand their purpose.



➤ Major Motherboard Ports:

Port / Connector	Purpose
24-pin ATX Power Connector	Supplies main power from the PSU to the motherboard.
8-pin CPU Power Connector	Provides power to the CPU.
SATA Ports	Connect SATA HDDs and SSDs to the motherboard.
USB Headers	Connect the front-panel USB ports of the computer case.
Front-Panel Header	Connects the power switch, reset switch, power LED and HDD LED.
Fan Headers	Provide power and control for CPU/case cooling fans.
Rear USB Ports	Connect external USB devices such as keyboards, mice and storage devices.
Audio Ports	Connect speakers, headphones and microphones.
LAN/Ethernet Port	Provides wired network/internet connectivity.

➤ Expansion Slots:

Slot	Purpose
PCIe x16 Slot:	Used mainly for a dedicated graphics card (GPU).
PCIe x1 Slots:	Used for smaller expansion cards such as sound, network or other interface cards.
M.2 Slot:	Used for compatible M.2 SSDs, including NVMe SSDs depending on motherboard support.

➤ CPU Socket & Compatible CPU Families

CPU Socket	Compatible CPU Families
LGA1151	supported 6 th /7 th /8 th /9 th generation Intel Core Processor.

❖ Task 3: RAM & Storage Installation

- Safely remove the RAM module, noting ESD precautions taken.
- Reinstall the RAM module correctly and confirm it is seated properly.
- Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).
- Confirm both components are detected correctly after reinstallation.

➤ **Ans:** - To safely remove and reinstall a RAM module and a storage device, then verify that both components are detected correctly.

➤ Step 1: Power Off and Follow ESD Precautions:

- I shut down the computer completely and disconnected the power cable before opening the case. I also discharged static electricity by touching a grounded metal part of the case and handled the components only by their edges.

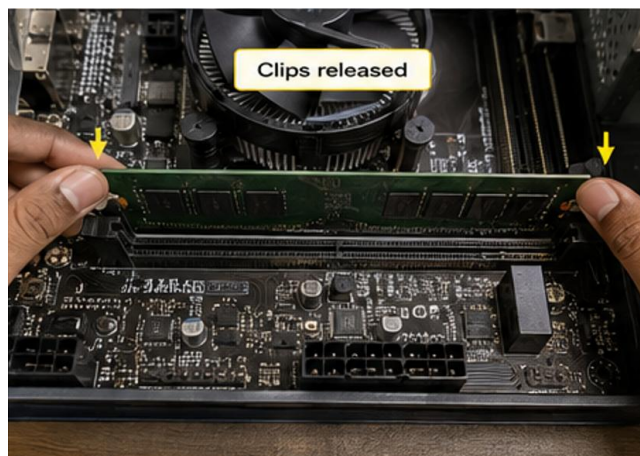


➤ I checked for:

- PC completely powered off
- Power cable disconnected
- Safe handling of RAM and storage components
- No unnecessary contact with component contacts

➤ Step 2: Remove the RAM Module:

- I located the RAM module on the motherboard and released the locking clips at both ends of the RAM slot. I carefully removed the RAM module by holding it from the edges.

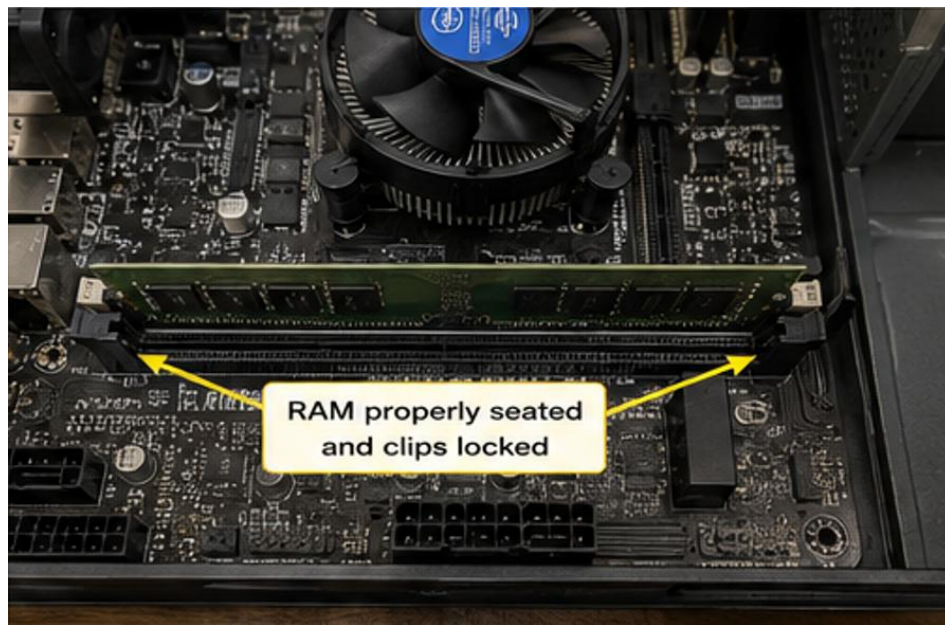


➤ I checked for:

- Correct RAM slot
- Locking clips released
- RAM removed without touching the gold contacts

➤ Step 3: Reinstall the RAM Module:

- I aligned the notch on the RAM module with the key in the RAM slot and pressed the module down evenly until the locking clips secured it.



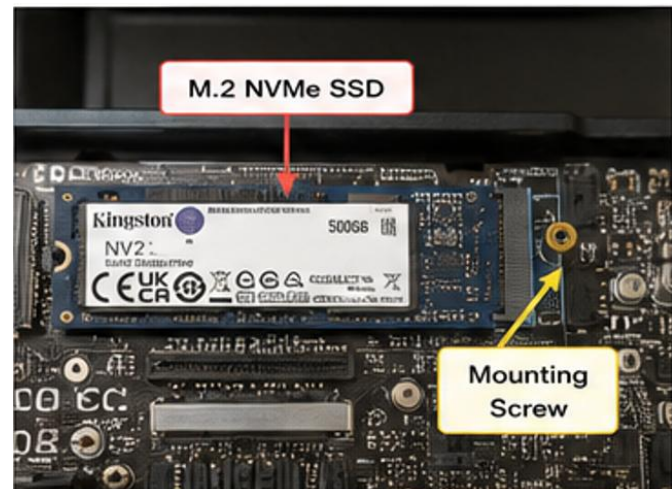
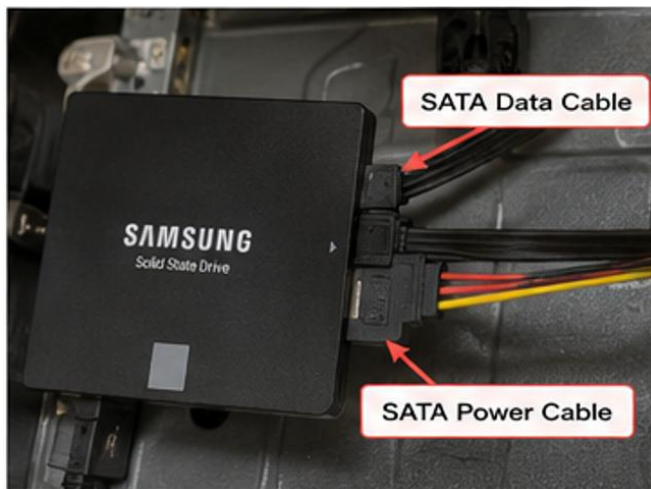
➤ I checked for:

- RAM notch correctly aligned
- Module fully inserted
- Locking clips completely closed

➤ Step 4: Remove and Reinstall the Storage Device:

- I identified the installed storage device and disconnected it carefully.
- For a SATA drive, I disconnected the SATA data and power connections.

- For an NVMe M.2 SSD, I removed the retaining screw, carefully lifted the SSD, and reinstalled it into the M.2 slot before securing it.

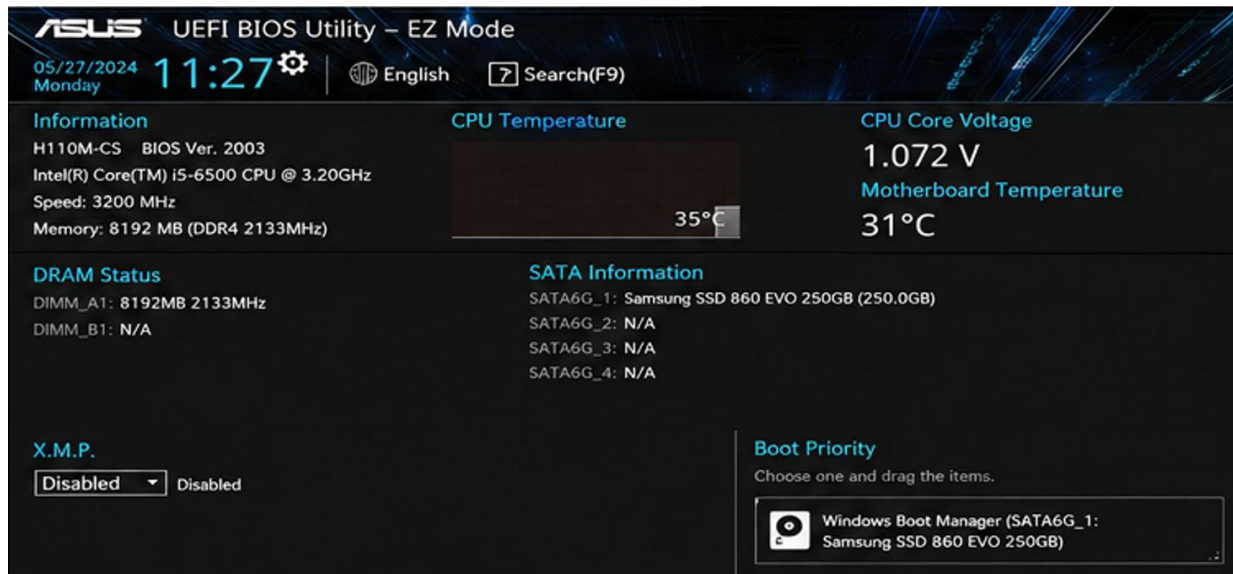


➤ I checked for:

- Correct storage connection type: SATA or NVMe
- Secure data/power connection for SATA
- Proper M.2 seating and retaining screw for NVMe

➤ Step 5: Power On and Confirm Detection:

- After reinstalling both components, I closed the case, reconnected power, and started the computer.
- I entered BIOS/UEFI and checked whether the RAM and storage device were detected correctly. I also allowed the operating system to boot and verified the storage device from the system.

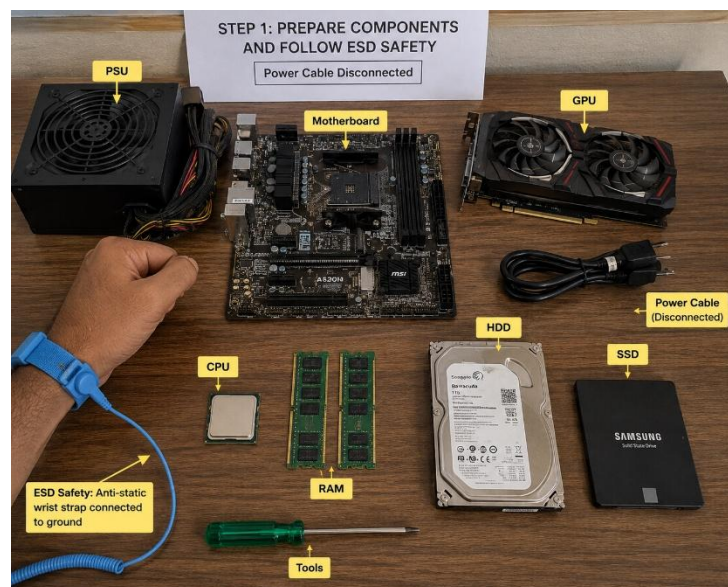


➤ I checked for:

- Correct RAM capacity detected
- Storage device detected
- System boots normally
- No hardware detection error

❖ Task 4: Full Desktop PC Assembly

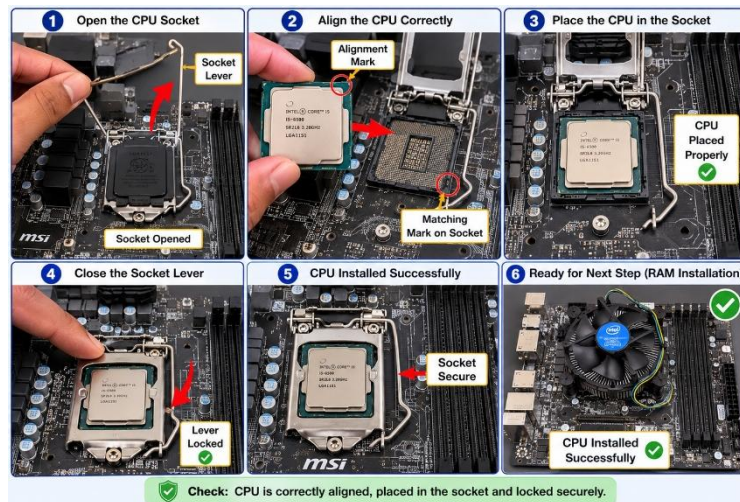
- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
 - Follow correct installation sequence and ESD/safety precautions throughout.
 - Document each step with labeled photos or diagrams.
 - Power on the system and confirm it passes POST successfully.
- To assemble a complete desktop PC using the provided components and verify that the system starts and successfully completes POST.
- Step 1: Prepare Components and Follow ESD Safety:
- I collected the motherboard, CPU, RAM, storage device, GPU, PSU and required tools. I switched off and disconnected power and followed basic ESD precautions before handling the components.
- **Check:** All components are present and safely handled.



➤ Step 2: Install the CPU:

- I carefully aligned the CPU with the marking/notch on the motherboard socket and installed it without forcing it. I then secured the CPU according to the socket mechanism.

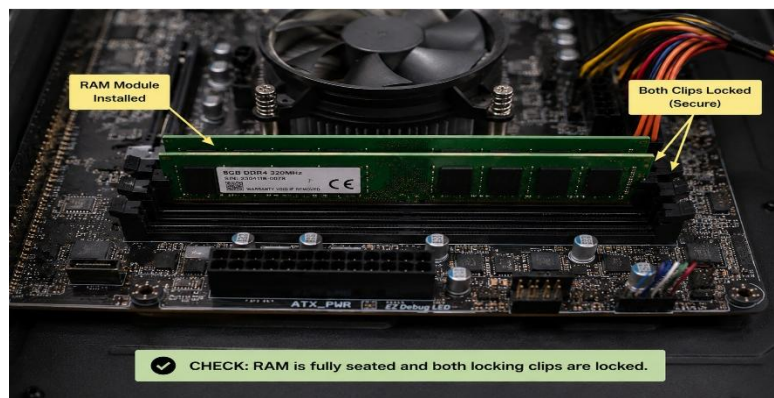
➤ **Check:** CPU is correctly aligned and securely installed.



➤ Step 3: Install the RAM:

- I aligned the RAM notch with the key in the DIMM slot and pressed the module down evenly until the locking clips secured it.

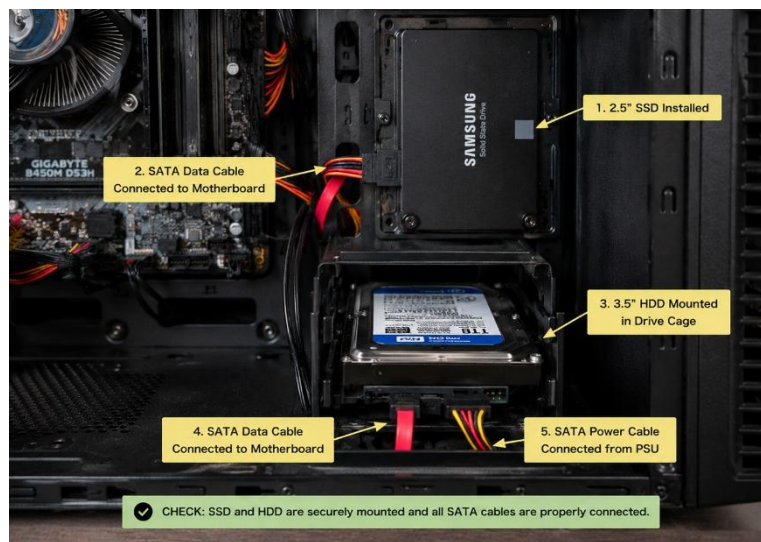
➤ **Check:** RAM is fully seated and both locking clips are locked.



➤ Step 4: Install the Storage Device:

- I installed the storage device using its supported connection.
- **SATA SSD/HDD:** connected SATA data and power cables.
- **NVMe SSD:** inserted the drive into the M.2 slot and secured it with the retaining screw.

➤ **Check:** Storage is firmly connected and correctly mounted.



➤ Step 5: Install the Motherboard in the Case:

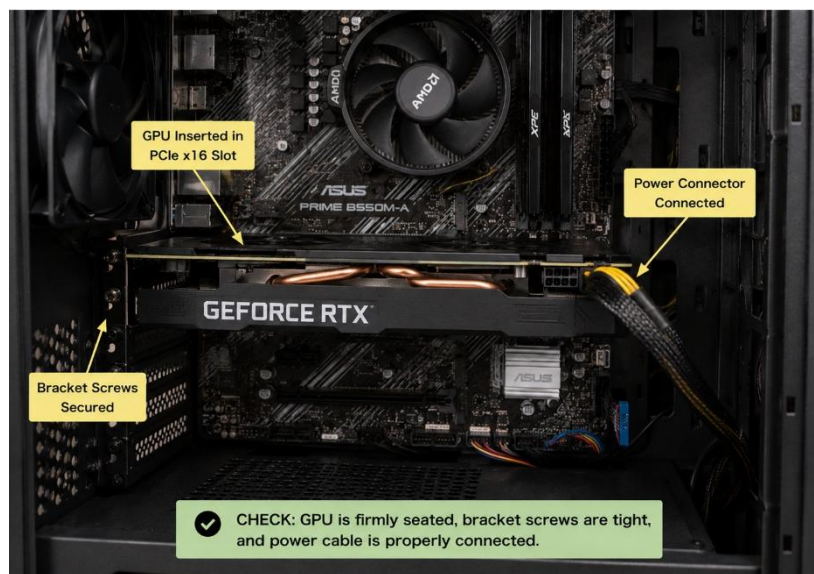
- I carefully placed the motherboard into the case, aligned it with the I/O opening and mounting points, and secured it using the appropriate screws.
- **Check:** Motherboard is securely mounted and no unnecessary contact is made with the case.



➤ Step 6: Install the GPU:

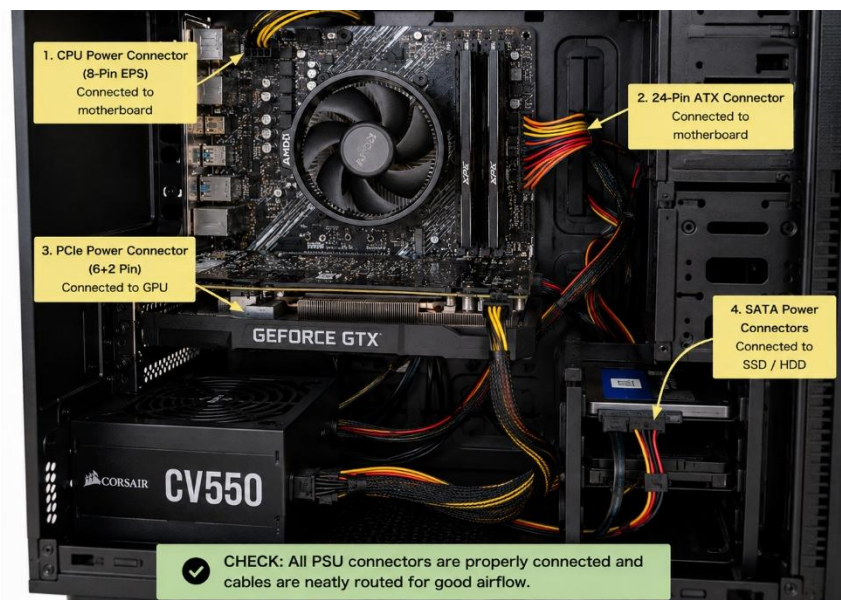
- I inserted the GPU into the appropriate PCIe x16 slot and secured it to the case. If required, I connected the GPU's PCIe power connector from the PSU.

➤ **Check:** GPU is fully seated and securely connected.



➤ Step 7: Connect the PSU:

- I connected the required PSU cables:
 - 24-pin ATX → Motherboard
 - CPU power connector → CPU/motherboard
 - PCIe power → GPU, if required
 - SATA power → SATA storage device, if required
- I ensured that all connectors were firmly seated.



➤ Step 8: Final Inspection and First Boot:

- Before powering on, I checked that:
- RAM and GPU were properly seated.
 - CPU cooler was installed.
 - Storage was connected.
 - PSU connectors were secure.

- No loose screws or cables were interfering with fans.
 - All necessary power connections were present.
- I then connected the power cable and switched on the PC.



- Step 9: Confirm Successful POST:
- I powered on the PC and observed the startup process.
- Successful POST was confirmed by:
- System powered on normally.
 - Display appeared on the monitor.
 - No abnormal POST error/beep was observed.
 - BIOS/UEFI screen was accessible.
 - CPU and RAM were detected.
 - Storage device was detected.



➤ Final Result:

- The desktop PC was assembled using the motherboard, CPU, RAM, storage, GPU and PSU. All major connections were checked, the system powered on successfully, and the hardware was detected during POST/BIOS.

[Section C — Mini Capstone Project]

❖ Workstation Deployment & Boot-Failure Diagnosis

❖ Objective:

- **ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.**

➤ **Ans: -** 1. First Three Components to Check When a Newly Assembled PC Fails to Boot.

➤ If a newly assembled PC does not boot, I would check the following three components first:

➤ **1. PSU and Power Connections:**

- I would first check the PSU because the system requires proper power to start. I would verify the PSU switch, AC power cable, 24-pin motherboard connector, CPU power connector, and GPU power connector if required.

➤ 2. RAM:

- I would check the RAM because incorrectly seated or incompatible memory can prevent the system from completing POST and may cause a blank display or error/beep codes. I would reseal the RAM and, if necessary, test one module at a time.

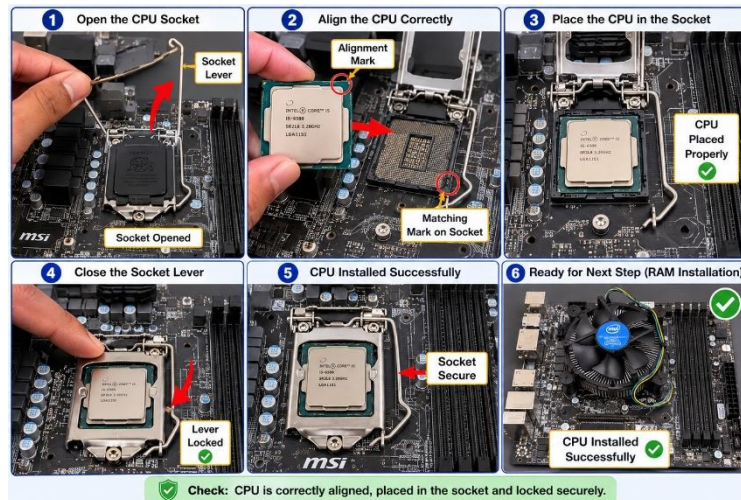
➤ 3. Motherboard and CPU:

- I would then inspect the motherboard and CPU because they are responsible for initializing the system during POST. I would check CPU installation, CPU power connection, motherboard mounting, and any visible signs of incorrect installation.

➤ This order starts with basic power and then checks the components most directly involved in the initial boot and POST process.

➤ 2. Motherboard–CPU Compatibility:

- Motherboard–CPU compatibility is important because the processor must match the motherboard's CPU socket, chipset support, BIOS support, and supported CPU generation.
- For the system I assembled/inspected, I would verify the exact motherboard model [**ASUS PRIME B550M-A**], CPU model [**AMD Ryzen 5 5600G**] socket type, and the manufacturer's supported CPU list. I would also confirm that the installed BIOS_version supports the selected processor.
- If the CPU is incompatible, the system may fail to boot, fail POST, show no display, or fail to detect the processor correctly.



➤ 3. Verifying RAM and Storage Installation:

- After installing the RAM and storage device, I would verify both physically and through BIOS/UEFI.

➤ RAM verification:

- Check that the RAM module is fully inserted into the correct DIMM slot.
- Confirm that the locking clips are securely closed.
- Enter BIOS/UEFI and verify that the correct RAM capacity is detected.
- Check that there are no POST memory errors or warning beeps.

➤ Storage verification:

- For a SATA drive, check both the SATA data cable and SATA power cable.
- For an NVMe SSD, check that the SSD is correctly seated in the M.2 slot and secured with its screw.

- Enter BIOS/UEFI and confirm that the storage device appears in the storage or boot-device list.
 - After the operating system starts, confirm that the drive is also visible to the OS.
- A successful BIOS/UEFI detection of the installed RAM and storage confirms that the components are properly connected and recognized by the system.



➤ **4. Hardware Installation Checklist:**

- The following checklist can help prevent similar hardware failures during future workstation deployments:

Checklist Item	Purpose
Follow ESD precautions before handling components.	Prevent static-electricity damage.
Verify CPU, motherboard and RAM compatibility before installation.	Prevent compatibility and boot problems.
Check all power and data connections before first boot.	Ensure every component receives the required power/data connection.
Confirm RAM and storage are properly seated.	Prevent hardware detection failures.
Enter BIOS/UEFI after first boot and verify CPU, RAM and storage detection.	Confirm successful hardware initialization.



➤ **5. Why SSD is Better Than HDD for Modern Office Workstations:**

- An SSD is more suitable than an HDD for modern office workstations because it provides faster boot times, quicker application loading, faster file access, and better overall system responsiveness.
- SSDs have no moving mechanical parts, so they are generally more resistant to physical shock and operate more quietly than HDDs. For the ABC Solutions computer lab, where users regularly work with office applications and need responsive systems, SSD storage would provide a better everyday user experience.
- Therefore, I would recommend SSD storage instead of HDD for modern office workstations because of its higher performance, faster access times, and better responsiveness.

➤ **Final Verification:**

- After completing the installation, I would power on the workstation and verify that it successfully completes POST, displays the BIOS/UEFI screen, detects the CPU and RAM, and recognizes the installed storage device. If all checks pass, the workstation can proceed to operating-system setup and deployment.

[Section D — AI-Augmented Learning]

❖ STEP 1 · BUILD WITH AI

❖ Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.

➤ 1. Exact Prompt Given to AI

- Prompt:
- Create a step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Arrange the checks in the correct priority order. The checklist must cover the motherboard, RAM, storage, and PSU as possible failure points. Include ESD/safety precautions and explain what the technician should check at each step. Apply the checklist to the ABC Solutions scenario from Section C and identify which step would most likely detect an incorrect installation or compatibility issue.

➤ **2. AI-Generated Original Troubleshooting Checklist**

- Desktop PC Fails to Boot — Initial Troubleshooting Checklist

➤ **Step 1: Check the PSU and Power Connections:**

- Confirm the power cable is connected.
- Check that the PSU switch is ON.
- Check the 24-pin motherboard power connector.
- Check the CPU power connector.
- Check GPU power connection if required.

➤ **Step 2: Check the RAM:**

- Make sure the RAM is properly seated.
- Check that the locking clips are closed.
- Reseat the RAM if necessary.
- Test one RAM module at a time if multiple modules are installed.

➤ **Step 3: Check the Motherboard and CPU:**

- Check whether the CPU is correctly installed.
- Verify that the CPU power connector is connected.
- Check for visible installation problems.
- Check motherboard indicators or POST error codes.

➤ **Step 4: Check the Storage Device:**

- Check SATA data and power connections for SATA drives.
- Check that an NVMe SSD is properly seated in its M.2 slot.
- Enter BIOS/UEFI and check whether the storage device is detected.

➤ Step 5: Check BIOS/UEFI and Compatibility:

- Enter BIOS/UEFI.
- Verify CPU and RAM detection.
- Verify storage detection.
- Check whether the motherboard BIOS supports the installed CPU.

➤ ABC Solutions Application:

- For the ABC Solutions scenario, the technician should first check the physical installation and connections. An incorrectly installed RAM module, loose power connection, incorrectly connected storage device, or incompatible CPU/motherboard combination could prevent the system from booting.

❖ **STEP 2 · TEST & DEBUG (WITHOUT AI)**

- ❖ **Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself.**

➤ 1. Corrected Troubleshooting Checklist:

- After reviewing the AI-generated checklist, I identified that ESD/safety precautions should be explicitly placed before physical troubleshooting, rather than being assumed. I also made the priority order clearer so that basic power and physical connections are checked before moving to BIOS or compatibility checks.

➤ **Corrected Desktop Boot-Failure Checklist**

➤ Step 1: Follow ESD and Safety Precautions:

- Shut down the system and disconnect AC power.
- Do not work on powered components.
- Discharge static electricity/use appropriate ESD protection.
- Handle RAM, CPU and other components by their edges.

➤ **Purpose:** Prevent accidental hardware damage and ensure safe troubleshooting.

➤ Step 2: Check PSU and Power Connections:

- Confirm the AC power cable is connected.
- Check the PSU switch.
- Verify the 24-pin motherboard connector.
- Verify the CPU power connector.
- Check GPU power if the GPU requires it.

➤ **Purpose:** A missing or loose power connection can prevent the PC from starting or completing POST.

➤ Step 3: Check RAM:

- Confirm that the RAM is installed in the correct slot.
- Check that both locking clips are secured.
- Reseat the RAM if necessary.
- If multiple modules are installed, test one module at a time.

➤ **Purpose:** Incorrectly seated or incompatible RAM can cause a blank display or POST failure.

➤ Step 4: Check Motherboard and CPU:

- Verify that the CPU is correctly seated.
- Check the CPU socket and installation.
- Confirm CPU power is connected.
- Check motherboard POST indicators or error codes.

➤ **Purpose:** The motherboard and CPU are responsible for initializing the system during POST.

➤ Step 5: Check Storage:

- For SATA drives, check SATA data and power connections.
- For NVMe SSDs, verify proper M.2 installation.
- Enter BIOS/UEFI and check whether the storage device is detected.

➤ **Purpose:** Confirms that the storage device is correctly installed and recognized.

➤ Step 6: Check BIOS/UEFI and Compatibility:

- Verify CPU detection.
- Verify RAM capacity.
- Verify storage detection.
- Check motherboard BIOS version/support for the installed CPU.

➤ **Purpose:** Identifies hardware compatibility or BIOS-related problems after the physical checks have passed.

➤ **Application to ABC Solutions:**

- For the ABC Solutions scenario, this corrected checklist would help identify incorrect installation or compatibility problems systematically. The most likely early detection points would be the power/RAM/CPU-motherboard checks, followed by BIOS/UEFI verification for hardware detection and compatibility.

➤ **2. What I Changed and Why:**

- I added ESD and safety precautions as the first step because hardware should be handled safely before troubleshooting.
- I clarified the priority order so that physical power and component connections are checked before BIOS/software-level checks.
- I also separated the motherboard/CPU and BIOS compatibility checks to make the troubleshooting process easier for a junior technician to follow.
- These changes make the checklist safer, more systematic, and easier to apply to the ABC Solutions boot-failure scenario.